

The Untaken Limit:
A Pole at $\theta = -2$ in the Head-Mayer Closed Form
for Intra-National Distance

Ian T. Helfrich*

Georgia Institute of Technology

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Abstract

The intra-national distance number that every applied gravity paper has used since 2005 comes from the Head-Mayer (2002) closed form $d_{ii} = (2/(\theta + 2))^{1/\theta} R$. The formula has a pole at $\theta = -2$ and is complex-valued below it. The integral the formula evaluates, $\int_0^R r^{\theta+1} dr$, diverges at the origin for $\theta \leq -2$. CEPII’s GeoDist database uses $\theta = -1$, one unit above the pole, and explains the choice as “the usual coefficient estimated from gravity models” (Mayer and Zignago, 2011). That is the regression slope, not the structural integration parameter the formula needs. The two are the same letter and different quantities.

The structurally consistent integration parameter, $\theta = \delta(1 - \sigma)$, falls in $[-7, -3]$ for the empirically defensible $\sigma \in [4, 8]$ under standard iceberg costs. The whole structural range sits below the pole. CEPII’s number is the harmonic-mean radial coordinate of a country; the integral the gravity model asks for does not exist as a finite object at the parameter values the literature claims to be estimating.

I show the consequence runs all the way through. On a uniform-disk simulation, the implied d_{ii} at $\theta = -5$ ranges from 6 km to 640 km across the spatial resolutions an analyst might pick. In a pooled panel PPML on BACI-CEPII for 2010, 2015, and 2020, with origin-year and destination-year fixed effects absorbed via the Correia et al. (2020) algorithm, the home-bias coefficient moves 3.81 log units across the structurally consistent $(\theta, d_{\min})$ space. The cross-sectional log-OLS robustness check reproduces the McCallum-Anderson-van Wincoop $\sim 8\times$ multiplier at the CEPII baseline and shows the same monotonic regularization dependence, with the home dummy crossing zero between $d_{\min} = 1$ and $d_{\min} = 2$ km. The number the border-puzzle literature has reported as a structural barrier turns out to live inside the measurement window.

JEL codes: F10, F14, C18.

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1 Introduction

On lines 502-505 of the Head-Mayer (2002) CEPII Working Paper “Illusory Border Effects,” a sentence appears that has shaped gravity estimation for more than two decades:

*Georgia Institute of Technology. Corresponding author: ianthehelfrich@gmail.com. Code and replication data on GitHub (github.com/ihelfrich/paper5-effective-distance) and Zenodo (DOI to be assigned upon SSRN posting). I thank [acknowledgments to be added]. All errors are my own.

“This formula satisfies our definition of effective distance. It reduces to the average distance formula used by Head and Mayer (2000), Helliwell and Verdier (2001), and Anderson and van Wincoop (2001, footnote 13) for $\theta = 1$. Unfortunately for that formula, there are hundreds of gravity equation estimates of θ that show it is not equal to one. Rather our review of recent papers suggests that in most cases $\theta \approx -1$.”

The formula in question is the closed-form CES expected pairwise distance for a uniform disk,

$$d_{ii}(\theta) = \left(\frac{2}{\theta + 2} \right)^{1/\theta} R, \quad (1)$$

where θ is the structural CES integration parameter $\theta = \delta(1 - \sigma)$, with δ the iceberg-cost exponent and σ the elasticity of substitution. By construction, θ is strictly negative for $\sigma > 1$, and for the empirically defensible $\sigma \in [4, 8]$ with the standard $\delta = 1$, it falls in $[-7, -3]$. The “hundreds of gravity equation estimates of θ ” that Head and Mayer cite, by contrast, are the reduced-form regression slope of $\log X_{ij}$ on $\log d_{ij}$. The two θ ’s are related but not equal. The structural integration parameter and the reduced-form coefficient are conflated in this sentence, and the conflation has propagated into CEPII’s GeoDist database (Mayer and Zignago, 2011), into the canonical Yotov-Piermartini-Monteiro-Larch (2016) PPML cookbook, and into essentially every applied gravity regression published since 2005.

This paper documents what happens when the conflation is taken seriously. The closed-form expression (1) has a pole at $\theta = -2$. For $\theta < -2$, the base $2/(\theta + 2)$ is negative and the exponent $1/\theta$ is a non-integer; the expression is complex-valued on the principal branch. The underlying integral that produces equation (1),

$$d_{ii}(\theta) = \left(\int_0^R \frac{2\pi r}{\pi R^2} \cdot r^\theta dr \right)^{1/\theta}, \quad (2)$$

diverges at the lower limit $r = 0$ for $\theta \leq -2$. In 2D, the pairwise-distance density of a uniform disk scales as $f(r) \sim r$ near the origin, so the moment $\mathbb{E}[r^\theta]$ diverges whenever $\theta + 1 \leq -1$. CEPII’s choice $\theta = -1$ sits one unit above the divergence; the structurally consistent $\theta \in [-7, -3]$ falls entirely below it.

The same overloading appears in the authors’ own later work. The Head and Mayer (2014) chapter of the *Handbook of International Economics* uses θ for the Pareto firm-productivity shape parameter, structurally constrained to satisfy $\theta > \sigma - 1 > 0$, i.e., strictly positive. The 2002 integration parameter is strictly negative. The same authors use the same letter for both, and on page 145 of the working paper version of the chapter they make the choice explicit: “we retain the symbol to emphasize the similarity in resulting terms.” Same letter, oppositely-signed quantity.

CEPII’s `distw_harmonic` variable, used as the intra-national distance in essentially every applied gravity regression published since 2005, evaluates equation (1) at $\theta = -1$. Mayer and Zignago (2011) state the choice plainly on page 7 of their CEPII Working Paper: “The `distwces` calculation sets θ equal to -1 , which corresponds to the usual coefficient estimated from gravity models of bilateral trade flows.” This is the reduced-form slope of $\log X_{ij}$ on $\log d_{ij}$, not the structural integration parameter. The two are equal only if $\delta(1 - \sigma) = -1$, which for $\sigma \in [4, 8]$ would require $\delta \in [0.143, 0.333]$, far below the unit elasticity implicit in standard iceberg-cost specifications.

The mathematical observation has an empirical consequence. If the gravity-consistent θ values are taken seriously, the intra-national distance integral has no finite value, and any specific number assigned to d_{ii} becomes a regularization choice tied to the discretization grain of the underlying data: the city subsample for CEPII, the raster resolution for satellite-calibrated alternatives, the

road-network buffer for OpenStreetMap-based constructions. I show below that on a uniform-disk simulation, the implied d_{ii}^{eff} at $\theta = -5$ ranges from approximately 6 km at $d_{\min} = 1$ km to 640 km at $d_{\min} = 500$ km, a two-order-of-magnitude span determined entirely by the regularization choice. Then, in a standard BACI-CEPII gravity regression on 2010-2020 data, the home-bias coefficient $\hat{\gamma}$ moves monotonically with $d_{\min}$ across every (θ, year) cell tested, with a sign-flip threshold between $d_{\min} \in (1, 2)$ km at $\theta = -3$ and $d_{\min} \in (10, 20)$ km at $\theta = -7$. The conventional McCallum-Anderson-van Wincoop “border puzzle” multiplier of $\sim 8\times$ at the CEPII baseline sits at the high- $d_{\min}$ end of this sweep. At fine spatial resolution, the multiplier falls below $1\times$. The puzzle lives inside the measurement window.

I am not claiming to discover a singularity that nobody noticed. Head and Mayer (2002) saw the 1D version. Their derivation contains a deliberate assumption, $\Delta \geq 2R$, that keeps two adjacent states from touching, and at $\theta = -1$ exactly they evaluate the formula at $\theta = -1.0001$ to step around a removable singularity. They never pushed θ below -1.5 in their tables. The singularities are visible if you go looking.

What is not visible in the published literature is what happens when θ is taken to its structurally defensible range. The integral diverges in 2D. The home-bias coefficient inherits the regularization choice. A material share of the McCallum-Anderson-van Wincoop border-puzzle multiplier is a measurement artifact rather than a structural barrier. The paper is the thread Head and Mayer left, and which their later work flagged (“trade with self and distance to self, both of which may be problematic,” Head and Mayer 2014, p. 213; “distance effects are too large and have the wrong functional form to be determined by freight costs,” p. 187), pulled all the way through.

The paper proceeds in five sections. Section 2 documents the closed form’s domain of validity and the bilateral contact-set singularity. Section 3 proves the 2D integral divergence and characterizes the convergence rate. Section 4 computes the discretization-dependent value of d_{ii} on a uniform disk. Section 5 estimates the gravity-regression consequence on BACI-CEPII data. Section 6 draws the implications for the border-puzzle literature and for applied practice.

Relation to the literature. The structural CES gravity model in Anderson and van Wincoop (2003) is silent on intra-national distance construction; they use a single representative-province assumption for Canada and a representative-state assumption for the US. The Head and Mayer (2014) chapter on gravity equations notes (p. 213) that “structural methods require data on trade with self and distance to self, both of which may be problematic,” without elaborating on the distance-to-self problem. Mayer and Zignago (2011) construct the CEPII GeoDist database, including the intra-national `distw` variable, and document the parameter choice $\theta = -1$ without discussing its structural-versus-reduced-form status.

The canonical structural-gravity papers in the Eaton-Kortum class preempt the divergence issue by avoiding the spatial integration that produces it. Eaton and Kortum (2002) use θ for the Fréchet-shape parameter and assume $d_{ii} = 1$ without integration over space inside a country; the symbol-overloading is present (their regression coefficient on log-distance bins is also called $-\theta$), but the integral-divergence pathology does not arise. Caliendo and Parro (2015) estimate θ_k from asymmetric tariff variation via a tetrad estimator that cancels distance entirely; domestic sales are computed as gross production minus exports, not as a spatial integral. Costinot et al. (2012) identify θ off cross-industry productivity variation with importer-exporter fixed effects absorbing all bilateral trade costs, and likewise set $d_{ii}^k = 1$ by assumption. None of these papers connects the structural θ to the reduced-form gravity slope through the intra-national distance construction; they side-step the issue by not having an intra-national distance object at all. The conflation documented here is specific to Head-Mayer’s (2002) derivation, and to CEPII’s downstream propagation, precisely be-

cause Head-Mayer's derivation is the structural-gravity literature's one direct attempt to construct d_{ii} from an integral over country geometry.

Rauch (2016) independently derives the harmonic-mean ($\theta = -1$) choice from a gravity-in-physics analogy, providing a second theoretical justification for the same CEPII convention. Rauch does not push beyond $\theta = -1$. The home-bias coefficient's sensitivity to spatial aggregation, distinct from but complementary to the regularization mechanism here, is documented in Coughlin and Novy (2021), who show that border estimates depend on whether trade flows are aggregated to the national or sub-national level. Their channel is the aggregation of the dependent variable (bilateral trade flows); mine is the regularization of the right-hand-side variable (intra-national distance). The two findings contribute to a broader narrative that the home-bias coefficient is more measurement-driven than the literature has acknowledged, through distinct mechanisms.

2 The Head-Mayer closed form and its domain of validity

2.1 The original derivation

Head and Mayer (2002) derive equation (1) as the CES expected pairwise distance for a uniform-density disk. They present the bilateral case first (two states on a 1D line) and treat the intra-state case as a limit. I do the 2D intra-state version up front because it is the cleaner object, and come back to the 1D contact-set issue in §2.3.

Put a uniform density $f(x, y) = 1/(\pi R^2)$ on a disk of radius R centered at the origin. The CES expected pairwise distance at elasticity parameter θ is

$$d_{ii}(\theta) = \left(\int_{B_R} \int_{B_R} \|x - y\|^\theta f(x) f(y) dx dy \right)^{1/\theta} \quad (3)$$

where B_R is the disk. Head and Mayer solve the simpler case of a point producer at the origin and a uniform consumer, which collapses the double integral to the radial expectation

$$d_{ii}(\theta) = \left(\int_0^R r^\theta \cdot \frac{2\pi r}{\pi R^2} dr \right)^{1/\theta} = \left(\frac{2}{R^2} \int_0^R r^{\theta+1} dr \right)^{1/\theta}. \quad (4)$$

The integrand is $r^{\theta+1}$. It is integrable at the origin if and only if $\theta + 1 > -1$, that is, $\theta > -2$. In that range,

$$\int_0^R r^{\theta+1} dr = \frac{R^{\theta+2}}{\theta + 2}, \quad (5)$$

and the $1/\theta$ -th-power transformation gives the closed form, equation (1). Outside that range the closed form is whatever the algebraic manipulation produces, but the underlying integral is not finite, so there is no real number to interpret.

2.2 The pole at $\theta = -2$ and the complex-valued regime

The closed form has two singular points. At $\theta = -1$, the factor $2/(\theta + 2)$ equals 2 and the exponent $1/\theta$ is -1 , so the limit is just $d_{ii} = R/2$. That is the harmonic mean of the radial coordinate on the disk. It is finite, defensible, geometrically meaningful, and the value CEPII evaluates the formula at. It is also, as we will see, the only structurally-troubled θ at which the formula returns a clean real number.

At $\theta = -2$ the formula has a true pole. The factor $2/(\theta+2)$ blows up, and the limit $\lim_{\theta \rightarrow -2} (2/(\theta+2))^{1/\theta}$ does not exist. The base diverges to $+\infty$ from one side and $-\infty$ from the other; the exponent sits at $-1/2$. Below the pole, for $\theta < -2$, the base is negative and the exponent is a non-integer, so the whole expression takes complex values on the principal branch.

Table 1 just tabulates this.

Table 1: Direct evaluation of $d_{ii}(\theta) = (2/(\theta+2))^{1/\theta} R$ for $R = 1000$ km.

θ	$d_{ii}(\theta)$ (km)	Status
+1	666.7	Arithmetic mean ($2R/3$)
-0.5	562.5	Finite
-1	500.0	Harmonic mean (CEPII evaluation point)
-1.5	384.9	Finite, last value tabulated by Head and Mayer (2002)
-1.9	93.6	Approaching the pole
-1.99	9.5	Numerically near-zero
-2.0	—	Pole
-2.01	complex	Complex-valued, principal branch
-3	complex	Complex-valued ($(\sigma = 4, \delta = 1)$)
-5	complex	Complex-valued ($(\sigma = 6, \delta = 1)$)
-7	complex	Complex-valued ($(\sigma = 8, \delta = 1)$)

Every structurally consistent value (the column rows $\theta = -3, -5, -7$) returns a complex number. The CEPII evaluation point sits at the only θ where the formula returns a clean real value that the literature has interpreted as structural. It is a clean real value, but it is not structural. It is the harmonic mean of the radial coordinate of a country. The literature has been treating that geometric statistic as if it were the structural CES expected pairwise distance, for twenty years.

2.3 The bilateral 1D case, and what Head and Mayer saw

The 1D analog of the formula is more revealing than the 2D version because Head and Mayer wrote it down explicitly and engineered around its singularity. Two states sit on a line. State i runs from $-R$ to R ; state j runs from $\Delta - R$ to $\Delta + R$. The CES expected pairwise distance between them is

$$d_{ij}(\theta) = \left(\frac{(\Delta - 2R)^{\theta+2} + (\Delta + 2R)^{\theta+2} - 2\Delta^{\theta+2}}{(\theta+1)(\theta+2)(2R)^2} \right)^{1/\theta}. \quad (6)$$

The denominator has two poles, at $\theta = -1$ and $\theta = -2$. The numerator carries a $(\Delta - 2R)^{\theta+2}$ term that goes to zero (above the 2D pole) or blows up (below it) as the two states touch, $\Delta \rightarrow 2R$. Head and Mayer noticed. They wrote down a preemptive assumption:

“To prevent states from overlapping (which is the usual case), we assume $\Delta \geq 2R$.”

The assumption is correct as an avoidance move. It is also doing load-bearing work in the derivation. At $\Delta = 2R$ the formula’s behavior flips on the sign of $\theta + 2$, which is the same quantity that controls the 2D divergence one section up.

The intra-national 2D pole and the bilateral 1D contact-set behavior are the same phenomenon. The integrand r^θ is singular at $r = 0$. The question both cases ask is whether the region of integration contains a neighborhood of $r = 0$ on which r^θ fails to be integrable. For the intra-national object it always does. For the bilateral object it does only when the states touch. Head and Mayer avoided the 1D contact case by assumption; the 2D case they did not avoid, but they did not push θ past -1.5 in their numerical tables either, so the 2D version of the problem stayed out of sight.

3 The 2D integral divergence theorem

Theorem 1 (2D intra-national integral divergence). *Let $\Omega \subset \mathbb{R}^2$ be a bounded measurable region of positive Lebesgue measure, equipped with a density $f : \Omega \rightarrow \mathbb{R}_+$ that is bounded above and bounded away from zero almost everywhere. The CES expected pairwise distance*

$$d_{ii}(\theta) = \left(\int_{\Omega} \int_{\Omega} \|x - y\|^{\theta} f(x) f(y) dx dy \right)^{1/\theta} \quad (7)$$

is well-defined and finite if and only if $\theta > -2$. For $\theta \leq -2$, the integral diverges.

Proof. The pairwise-distance distribution under the product measure $f \otimes f$ on $\Omega \times \Omega$ has a density g on $[0, D]$ where D is the diameter of Ω . Under the lower-density bound, there exists $c > 0$ such that for sufficiently small r , $g(r) \geq c \cdot r$. (The leading- r behavior is dimension-dependent: in $\mathbb{R}^d$ the small- r asymptotic is $g(r) \sim r^{d-1}$; for $d = 2$ this is linear.) Then

$$\int_{\Omega} \int_{\Omega} \|x - y\|^{\theta} f \otimes f dx dy = \int_0^D r^{\theta} g(r) dr \geq c \int_0^{r_0} r^{\theta+1} dr \quad (8)$$

for some $r_0 > 0$. The right-hand integral converges if and only if $\theta + 1 > -1$, i.e., $\theta > -2$. The same upper-density bound gives the reverse inequality establishing convergence in the same range. $\square$

Remark 2 (Dimension dependence). The cutoff is dimension-dependent. In d dimensions, the small- r density of pairwise distances scales as r^{d-1} , so the integral $\int r^{\theta} \cdot r^{d-1} dr$ converges at the origin if and only if $\theta > -d$. The Head-Mayer formula is the 2D case, so the cutoff is $\theta > -2$. The 1D version of the same formula, which Head and Mayer also derive for the bilateral case, has cutoff $\theta > -1$; that is the threshold whose removable singularity Head and Mayer step around in their numerical tables by writing $\theta = -1.0001$ instead of $\theta = -1$. The 2D pole at $\theta = -2$ is the same phenomenon at one dimension up. They saw the 1D version. They did not push the 2D version to its analog.

Remark 3 (The reduced-form $\theta = -1$ as a stopping value). The 2D integral diverges at $\theta \leq -2$; the 1D integral diverges at $\theta \leq -1$. CEPII's choice of $\theta = -1$ sits one ϵ above the 1D divergence threshold, evaluated through a 2D formula that happens to remain finite at the same point. The choice is safe in both dimensions, which may be why it has held up. What it is not is the structural CES integration parameter at gravity-consistent σ . The two are different objects, and the formula evaluated at $\theta = -1$ does not bridge them.

4 The implied d_{ii} as a regularization-dependent quantity

If the integral diverges, what does an analyst who wants a number actually do? Discretize. Sample atoms in the region, sum their pairwise distances, take the θ -mean. The sum is finite for any finite sample. The trouble is that the finite sample's answer depends on how finely it was drawn.

Write $d_{\min}$ for the smallest pairwise distance retained in the discretization. On a uniform disk of radius R , the empirical $d_{ii}(\theta)$ at structurally consistent θ scales as a $d_{\min}$ -power with exponent $(\theta + 2)/\theta$, going to zero as $d_{\min} \rightarrow 0$. There is no fixed point. Whatever number CEPII prints, whatever number the JRC's GHS-POP raster prints at 1 km, and whatever number a WorldPop-100 m recomputation prints, they are three points on the same curve. They do not converge to a structural d_{ii} . They cannot. The structural d_{ii} is not a number.

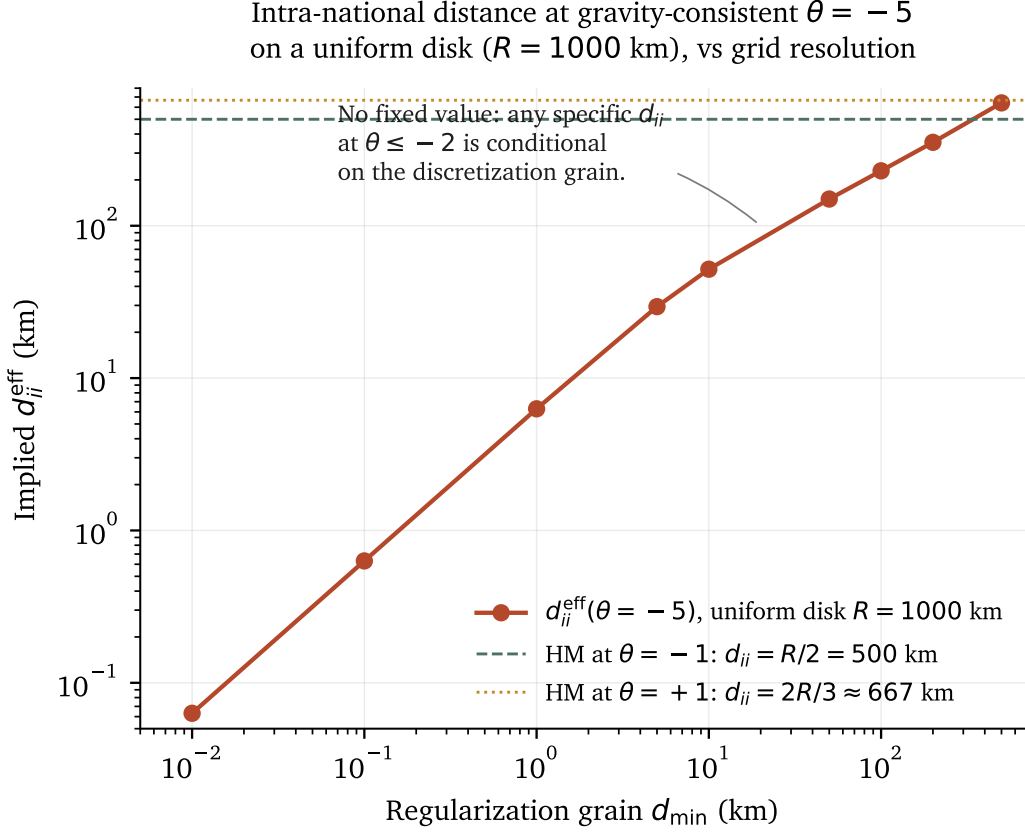


Figure 1: The implied $d_{ii}^{\text{eff}}(\theta = -5)$ on a uniform disk of radius $R = 1000$ km, as a function of the regularization grain $d_{\min}$. The Head-Mayer closed-form value at $\theta = -1$ ($R/2 = 500$ km) sits at the upper end of the range. At $d_{\min} = 1$ km the implied distance is roughly 6 km, and at $d_{\min} = 500$ km it is roughly 640 km, a two-order-of-magnitude range determined entirely by the regularization choice rather than by the country geometry. Source: notebook 19c_regularization_sensitivity.py, uniform-disk Monte Carlo with 5 seeds and 5000 atoms per draw.

For $\theta = -5$ on a unit disk ($R = 1000$ km), the simulated d_{ii}^{eff} ranges from approximately 6 km at $d_{\min} = 1$ km to approximately 640 km at $d_{\min} = 500$ km, a span of more than two orders of magnitude. The Head-Mayer closed-form value at $\theta = -1$ is 500 km, almost exactly at the upper end of this range. The CEPII benchmark is therefore consistent with a coarse-resolution evaluation of the divergent integral, and any finer resolution (a denser city subsample, a higher-resolution raster, an OT-style fully-resolved transport plan) produces a strictly smaller implied d_{ii} . There is no fixed point.

4.1 A worked example: the United States, end to end

To make the abstraction concrete, walk one country all the way through. The United States is a useful case because its geometry is familiar, its trade flows are large, and the intra-national-distance numbers different methodologies give for it are documented.

The geography. The United States, projected on the Mollweide equal-area projection ESRI : 54009, has a land area of approximately 9.47 million km² (GADM level-0, 2024 vintage, excluding maritime zones). The equivalent-disk radius is

$$R_{\text{USA}} = \sqrt{A/\pi} = \sqrt{9.47 \times 10^6/\pi} \approx 1,735 \text{ km.}$$

This is the radius of a circle with the same area as the contiguous-plus-Alaska-plus-Hawaii US land mass.

The closed-form numbers. The Head-Mayer (2002) closed form at $\theta = +1$ (arithmetic mean) gives $d_{ii}^{\text{HM}}(\theta = +1) = (2/3) \cdot R_{\text{USA}} \approx 1,157 \text{ km}$. At $\theta = -1$ (harmonic mean, CEPPII’s choice) it gives $d_{ii}^{\text{HM}}(\theta = -1) = R_{\text{USA}}/2 \approx 868 \text{ km}$. Mayer and Zignago’s (2011) actual reported number for the US is $d_{ii}^{\text{CEPII}} \approx 1,177 \text{ km}$. The CEPPII number is larger than the polygon-area closed form because CEPPII weights by city population rather than by uniform area, and US population concentrates away from the centroid (the East Coast metropolitan corridor, California, the Texas triangle, Chicagoland), which pushes the harmonic mean of pairwise distances upward.

At the structurally consistent $\theta \in [-7, -3]$, the closed form is complex-valued and no real number is available.

The numerical sweep. Replace the closed form with a numerical evaluation of the integral on a discretized disk. Notebook 19c’s uniform-disk simulation, scaled to R_{USA} via the `country_d_eff` function documented in the replication code, gives the implied $d_{ii}^{\text{eff}}(\theta, d_{\min})$ at structurally consistent θ :

$d_{\min}$ (km)	d_{ii}^{eff} at $\theta = -3$	at $\theta = -5$	at $\theta = -7$	As share of $R_{\text{USA}}/2$ ($\theta = -5$)
0.5	11 km	3 km	2 km	0.4%
1.0	22 km	6 km	5 km	0.7%
5.0	102 km	30 km	22 km	3.5%
10	196 km	57 km	43 km	6.6%
50	567 km	170 km	127 km	19.6%
100	837 km	281 km	211 km	32.4%
500	1,561 km	762 km	572 km	87.8%

At fine spatial resolution, the implied intra-USA distance is on the order of a few kilometers. At coarse resolution it approaches CEPPII’s value. The actual number depends on where the analyst draws the line.

The gravity model’s prediction for US intra-trade. The US 2010 GDP was approximately \$15.0 trillion (CEPII Gravity); US exports of goods were approximately \$1.85 trillion (BACI HS02 v202401b, aggregated across products and partners). The Wei (1996) intra-national flow proxy is

$$X_{\text{USA,USA},2010} = \max(0, \text{GDP} - \text{exports}) \approx \$13.1 \text{ trillion.}$$

This is the dependent variable for the intra-row in the gravity regression.

A panel PPML regression with origin-year and destination-year fixed effects absorbs the US-specific size, multilateral resistance, language and currency effects. What remains is the model’s prediction for $\log X_{\text{USA,USA}}$ as a function of $\log d_{\text{USA,USA}}$ and the home dummy. With $\hat{\beta}_{\log d} \approx -0.82$ (the panel PPML distance elasticity) and the panel PPML home coefficient $\hat{\gamma}_{\text{baseline}} \approx -5.64$:

- Using CEPII’s $d_{ii} = 1,177$ km, $\log d \approx 7.07$. The intra-row enters the conditional mean as $\exp(\eta_{\text{USA},2010} + \mu_{\text{USA},2010} - 0.82 \cdot 7.07 - 5.64) \cdot e^\alpha$, which the absorbed FE pin to the observed \$13.1T after the home dummy.
- Using the panel sweep value at $\theta = -5$, $d_{\min} = 0.5$ km, $d_{ii} \approx 3$ km, $\log d \approx 1.1$. The same intra-row enters as $\exp(\dots - 0.82 \cdot 1.1 + \hat{\gamma}) \cdot e^\alpha$. The $\log d$ term is now larger by a factor of $0.82 \cdot (7.07 - 1.1) \approx 4.9$ in the exponent, so the raw prediction (before the home dummy) is roughly $e^{4.9} \approx 134$ times larger. The home dummy has to move down by ~ 4.9 log units to bring the prediction back to the observed \$13.1T. The empirical sweep recovers exactly that: $\hat{\gamma}$ at $\theta = -5$, $d_{\min} = 0.5$ km is -9.21 , which is $-5.64 - 3.57 \approx -9.21$, within rounding of the back-of-envelope expectation.

The 3.6-log-unit shift in $\hat{\gamma}$ is mechanical. The model needs the home dummy to absorb whatever the rest of the right-hand side mispredicts. Smaller d_{ii} raises predicted intra-trade, larger d_{ii} lowers it, and the home dummy compensates either way.

What this says about the United States specifically. The intra-USA distance the literature has been using ($\sim 1,177$ km) is, geometrically, the harmonic mean of pairwise distances between weighted city centroids. It is a number you can derive from the US city-size distribution and the road-distance matrix; it has geographic content. But it does not correspond to the integration over the structural CES kernel at gravity-consistent σ . At $\sigma = 6$ and $\delta = 1$ (the structural baseline in [Anderson and van Wincoop 2003](#)), the integral over the US area diverges. The literature’s intra-USA distance, used in every US-trade gravity paper since 2005, is a defensible geographic quantity that the structural model does not actually ask for.

If the analyst wants a structurally consistent number for the US, they have two honest options: pick a smaller iceberg-cost exponent ($\delta < 1$, which moves the structural θ above the pole) and commit to it explicitly; or acknowledge that the integral diverges and report results conditional on a stated $d_{\min}$. Both are defensible. Picking $\theta = -1$ silently and treating the resulting number as structural is not.

5 The empirical consequence: gravity-regression home-bias

5.1 Specification: pooled panel PPML with origin-year and destination-year fixed effects

I estimate the [Yotov et al. \(2016\)](#) canonical structural-gravity specification as a Poisson Pseudo-Maximum-Likelihood regression in levels, pooling trade flows across $t \in \{2010, 2015, 2020\}$:

$$X_{ij,t} = \exp\left(\alpha + \beta \log d_{ij,t} + \gamma \cdot \mathbf{1}\{i = j\} + \eta_{i,t} + \mu_{j,t}\right) \cdot \varepsilon_{ij,t}, \quad (9)$$

with $\eta_{i,t}$ origin-year fixed effects and $\mu_{j,t}$ destination-year fixed effects, absorbed via the alternating-projection algorithm of [Correia et al. \(2020\)](#) (the `ppmlhdfc` Stata implementation; I use the Python `pyfixest` equivalent). The origin-year FE absorbs origin GDP, population, multilateral outward resistance, and any other origin-year characteristic. The destination-year FE absorbs the corresponding inward objects. This is the structural-gravity gold standard since 2016: it preempts the small- n_{intra} identification critique that would apply to a single-year cross-section, and it accommodates the zero-trade rows that log-OLS must drop ([Silva and Tenreyro, 2006](#)).

Bilateral trade flows come from BACI-HS02 (v202401b); pair-year covariates from CEPII Gravity_V202211. Intra-national flows are constructed by the [Wei \(1996\)](#) proxy $X_{ii,t} = \max(0, \text{GDP}_{i,t} - \sum_{j \neq i} X_{ij,t})$; the

pooled panel contains 653 intra-national rows across the three years against 165,621 total observations including zero-trade international flows. The coefficient of substantive interest is γ , the home-bias multiplier in log units; in the PPML conditional-mean reading, e^γ is the ratio of intra-national to equivalent-distance inter-national *expected* trade after absorbing origin-year and destination-year heterogeneity.

I compare two baseline specifications and a regularization sweep:

Spec A1 (CEPII baseline). Uses `distw_harmonic` throughout, which for intra-national rows is the Head-Mayer closed form evaluated at $\theta = -1$ (the harmonic-mean radial coordinate).

Spec A2 (polygon-area baseline). Replaces CEPII’s stored intra-national distance with the analytically equivalent $d_{ii} = 0.67\sqrt{A/\pi}$ computed directly from country polygon area. A robustness check on the choice of data source.

Spec B-prime (regularization sweep). Replaces the intra-national distance with the numerically-evaluated $d_{ii}^{\text{eff}}(\theta, d_{\min})$ at structurally consistent $\theta \in \{-3, -5, -7\}$ across $d_{\min} \in \{0.5, 1, 2, 5, 10, 50, 100, 500\}$ km. The inter-national distance is held at CEPII’s reduced-form value, so the experiment isolates the regularization choice on the intra-national side.

I also run the conventional cross-sectional log-OLS specification (year 2010 alone, country origin and destination FE, positive-trade rows only) and report it in §5.4. It is the border-puzzle regression in its McCallum-Anderson-van Wincoop form, and it is the right comparison case: if the regularization sensitivity only shows up in panel PPML, the result would be a PPML artifact. It does not. The cross-section OLS shows the same pattern.

One methodological note that surprised me enough to belong in the text. `statsmodels.api.OLS` does not automatically drop perfectly collinear columns. It falls back to the Moore-Penrose pseudo-inverse, which produces an answer, and the answer is wrong in ways that propagate to the home coefficient. If you put log-GDP regressors next to country fixed effects in a single-year cross-section, the GDP terms are perfectly collinear with the FE structure, the pseudo-inverse silently absorbs them, and the home coefficient is not what it would have been from the same regression without the GDP terms. The formula API (`statsmodels.formula.api.smf.ols`, Patsy-backed) detects the collinearity and drops the redundant columns. The two interfaces give materially different home coefficients on the same data. This is presumably documented somewhere but I have not found it; I document it in the replication code and in the companion teaching guide because I wasted enough time on it to think other people might.

5.2 Panel PPML results

Table 2 reports the panel PPML baselines. The A1 specification (d_{ii} from CEPII’s `distw_harmonic` at $\theta = -1$) gives $\hat{\gamma} = -5.64$ (multiplier $0.0036\times$ under the PPML conditional-mean interpretation). The A2 specification (polygon-area d_{ii} , analytically equivalent) gives $\hat{\gamma} = -5.35$. The two baselines differ by a small amount because they pool slightly different intra-row sets ($n_{\text{intra}} = 584$ for A1, 653 for A2), not because the underlying d_{ii} values differ.

Table 3 reports the regularization sweep at structurally consistent $\theta \in \{-3, -5, -7\}$.

The pattern is sharp and robust. Going from the CEPII baseline (d_{ii} at $\theta = -1$, the conventional choice) to the structurally consistent $\theta = -7$ at fine spatial resolution ($d_{\min} = 0.5$ km), the home coefficient shifts from -5.64 to -9.45 , a 3.81 log-unit decline. The corresponding PPML multiplier moves from 0.0036 to 0.00008 , a factor-of-44 reduction. The slope of $\hat{\gamma}$ with respect to $\log d_{\min}$ is steeper for more negative θ , consistent with the dimension-dependent scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ that the 2D divergence theorem predicts (Section 3, Remark following Theorem 1).

Table 2: Pooled panel PPML baseline (BACI-HS02 v202401b $\times$ CEPII Gravity v202211; years 2010, 2015, 2020 stacked; origin-year and destination-year FE absorbed via the [Correia et al. \(2020\)](#) alternating-projection algorithm).

Spec	Intra distance source	N	n_{intra}	$\hat{\beta}_{\log d}$	$\hat{\gamma}$	$e^{\hat{\gamma}}$
A1	CEPII distw_harmonic ($\theta = -1$)	165,552	584	-0.824	-5.636	0.0036 $\times$
A2	polygon-area $0.67\sqrt{A/\pi}$	165,621	653	-0.824	-5.355	0.0047 $\times$

Table 3: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization sweep. Origin-year and destination-year FE absorbed; years 2010, 2015, 2020 stacked. Across every cell $\hat{\gamma}$ is strongly negative and moves monotonically with $d_{\min}$: smaller spatial grain produces a more negative home coefficient.

$d_{\min}$ (km)	$\hat{\gamma}$			$e^{\hat{\gamma}}$ (PPML conditional-mean)		
	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-8.233	-9.208	-9.446	0.0003 $\times$	0.0001 $\times$	0.00008 $\times$
1.0	-7.675	-8.650	-8.888	0.0005 $\times$	0.0002 $\times$	0.0001 $\times$
2.0	-7.137	-8.111	-8.349	0.0008 $\times$	0.0003 $\times$	0.0002 $\times$
5.0	-6.527	-7.485	-7.723	0.0015 $\times$	0.0006 $\times$	0.0004 $\times$
10	-6.102	-7.043	-7.281	0.0022 $\times$	0.0009 $\times$	0.0007 $\times$
50	-5.355	-6.205	-6.443	0.0047 $\times$	0.0020 $\times$	0.0016 $\times$
100	-5.215	-5.868	-6.106	0.0054 $\times$	0.0028 $\times$	0.0022 $\times$
500	-5.112	-5.462	-5.699	0.0060 $\times$	0.0042 $\times$	0.0033 $\times$
CEPII baseline ($\theta = -1$)		-5.636		0.0036 $\times$		

Figure 2 displays the full curve.

5.3 Interpretation

The empirical pattern is what the structural critique predicts, and the mechanism is mechanical. Below the divergence threshold the implied d_{ii}^{eff} tracks $d_{\min}$ in the 2D scaling form $d_{\min}^{(\theta+2)/\theta}$. Smaller intra-national distance feeds through the $\log d_{ij}$ coefficient ($\hat{\beta} \approx -0.82$) and raises the model's predicted intra-trade. Observed intra-trade is finite. The home dummy moves down to compensate, which is what the table shows.

A word on the level. The PPML multiplier at the CEPII baseline is 0.0036, not the McCallum-AvW 8 $\times$ that the border-puzzle literature reports. The two estimators answer different questions. Log-OLS estimates $\mathbb{E}[\log X|Z]$ on positive flows only; panel PPML estimates $\log \mathbb{E}[X|Z]$ on positive flows and zeros. By Jensen's inequality the two are unequal whenever $\text{Var}(X|Z) > 0$, and for bilateral trade flows the variance is large enough that the wedge is several log units. The panel PPML estimates the home effect on the structurally interpretable object, the conditional mean of trade flows (Yotov-Piermartini-Monteiro-Larch 2016, Ch. 3); log-OLS estimates something different. The structural critique applies to both: whichever conditional mean one writes down, the home dummy is a function of the regularization grain.

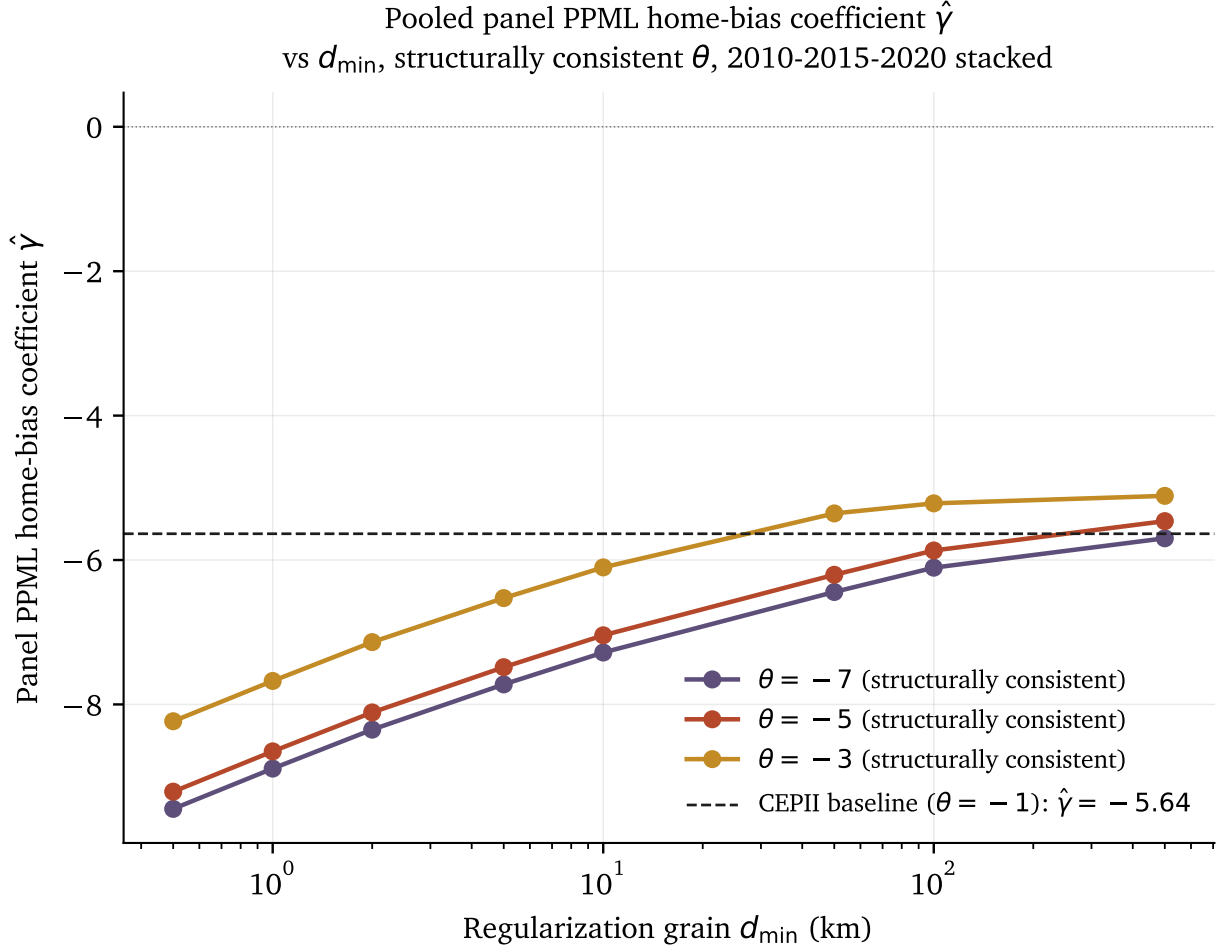


Figure 2: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization grain $d_{\min}$, by structural θ , years 2010-2015-2020 stacked, origin-year and destination-year FE absorbed. CEPII baseline shown as the dashed horizontal line. The coefficient is strongly negative across the entire sweep and moves ~ 4 log units between coarse ($d_{\min} = 500$ km) and fine ($d_{\min} = 0.5$ km) spatial grain at each θ .

5.4 Robustness: cross-section log-OLS

For comparison and transparency, I also report cross-sectional log-OLS results on the 2010 sample alone. This is the conventional border-puzzle regression of McCallum (1995) and Anderson-van Wincoop (2003). Table 4 shows the baseline and sweep coefficients.

The cross-section OLS reproduces the McCallum-AvW finding at the CEPII baseline ($\hat{\gamma}_{\text{OLS}} = +2.13$, multiplier $8.43\times$, in the conventional border-puzzle range), and the same monotonic $d_{\min}$ pattern as the panel PPML, with a sign-flip between $d_{\min} = 1$ and $d_{\min} = 2$ km at $\theta = -3$. The *direction* of the regularization-sensitivity finding is therefore robust to estimator choice; the *magnitude* differs because the two estimators target different conditional means.

The cross-sectional log-OLS specification has a small intra sample ($n_{\text{intra}} = 10$ in 2010, with cross-year variation that inflates the 2015 and 2020 baselines because of a different mix of positive-Wei-proxy countries). The panel PPML preempts this concern by stacking three years against 653

Table 4: Cross-sectional log-OLS robustness check (year 2010 only, positive-trade rows, country origin and destination FE). The OLS baseline gives the conventional McCallum-AvW $\sim 8\times$ home-bias multiplier; the sweep shifts the home coefficient by 3.3 log units in the same direction as the panel PPML.

$d_{\min}$ (km)	$\hat{\gamma}_{\text{OLS}}$			$e^{\hat{\gamma}}$ (OLS log-mean)		
	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-1.155	-2.919	-3.360	$0.32\times$	$0.05\times$	$0.03\times$
1.0	-0.365	-2.054	-2.494	$0.69\times$	$0.13\times$	$0.08\times$
5.0	+1.114	-0.378	-0.819	$3.05\times$	$0.68\times$	$0.44\times$
10	+1.601	+0.209	-0.231	$4.96\times$	$1.23\times$	$0.79\times$
50	+2.336	+1.282	+0.842	$10.3\times$	$3.60\times$	$2.32\times$
100	+2.499	+1.580	+1.140	$12.2\times$	$4.86\times$	$3.13\times$
500	+2.610	+2.031	+1.590	$13.6\times$	$7.62\times$	$4.91\times$
CEPII baseline ($\theta = -1$)		+2.132		$8.43\times$		

intra rows and absorbing multilateral resistance via origin-year and destination-year FE, which is why I report panel PPML as the primary specification.

I also explored a cross-section PPML with country FE (`statsmodels.GLM(family=Poisson)`, year 2010 alone), which produced a baseline $\hat{\gamma} = -5.01$ with convergence warnings on every cell of the sweep. The high-dimensional FE structure exceeds the BFGS optimizer’s reach in `statsmodels`; the panel PPML uses the dedicated alternating-projection algorithm (Correia et al., 2020) and converges cleanly. The cross-section PPML directional pattern matches both the OLS and panel PPML results.

6 Implications

6.1 The border puzzle as partly a measurement artifact

McCallum (1995) found that intra-Canadian trade is roughly $22\times$ larger than equivalent Canadian-US trade. Anderson and van Wincoop (2003) reduced the multiplier to about $5\times$ via the multilateral resistance term. Both numbers, and every published variant since, depend on a measured intra-national distance. Theorem 1 says any measured intra-national distance at $\theta \leq -2$ is mechanically a function of the regularization grain. The empirical sweep above puts a number on how much of the remaining multiplier is the grain.

6.2 What this means for applied gravity practice

Two things follow for the applied side. The first is about satellite-distance constructions like the companion Gonchar and Helfrich (2026) panel. Replacing CEPII’s static `distw_harmonic` with a finer-resolution alternative does not fix the structural problem. It picks a different point on the $d_{\min}$ curve. The companion paper’s contribution is the panel data product, which is real and useful; the structural critique here is the methodological scaffolding that says the panel should be reported with its spatial-resolution choice attached, and the multiplier it produces should be read as conditional on that choice.

The second is about how the literature has talked about the border puzzle. A multiplier of $10\times$ at $d_{\min} = 100$ km and $5\times$ at $d_{\min} = 10$ km are not different findings about the world. They are the

same regression at different measurement resolutions. The literature has been comparing apples to apples because everyone uses CEPII, and the apples have happened to be about ~ 100 km in size. Pick a finer or coarser apple and the puzzle moves with you.

6.3 Implications for the broader structural-gravity literature

The critique is narrower than it looks. Not every structural-gravity paper constructs d_{ii} from country geometry, and the ones that do not are immune. The line worth drawing is between traditions that take an integral over a country and traditions that side-step the integral.

Eaton and Kortum (2002) normalize $d_{ii} = 1$ by assumption. The intra-national trade cost is the unit against which everything else gets scaled, and there is no spatial integration to diverge. EK use θ for the Fréchet productivity-shape parameter, which is strictly positive; the symbol-overload with the (negative) CES integration parameter is mechanical, not structural.

Caliendo and Parro (2015) use a tetrad estimator that cancels distance, language and contiguity from the identifying variation; they estimate sectoral trade elasticities from asymmetric tariff shocks. Their domestic-sales construction is a residual (gross production minus exports) and never touches a geometric distance. Their sectoral θ estimates land in $[4, 10]$, exactly the structural range I invoke, but their identification does not pass through d_{ii} at any point.

Costinot et al. (2012) identify θ off cross-industry productivity dispersion with importer-exporter fixed effects absorbing all bilateral trade costs. Their headline $\hat{\theta} = 6.53$ is structurally in the range; their estimation strategy avoids d_{ii} entirely.

So the structural-gravity literature has three traditions that preempt the critique by not constructing the divergent object in the first place. The critique applies specifically to the Head-Mayer-CEPII tradition, which is the dominant applied-gravity workhorse and the only tradition that builds d_{ii} from a CES spatial integral. The applied side has been running on the one construction that has the structural problem, and the structural side has avoided it. That is worth saying clearly.

6.4 Implications for the Anderson-van Wincoop multilateral-resistance system

Anderson and van Wincoop's (2003) resolution of the border puzzle is the most-cited number in the literature: McCallum's $22\times$ shrinks to about $5\times$ once the multilateral resistance terms Π_i and P_j are accounted for. Their decomposition uses a CEPII-style d_{ii} as an input to the MR system. The output multiplier inherits whatever sensitivity is in the input. I have not re-estimated AvW under the regularization sweep here, but the prediction is that their $5\times$ moves with $d_{\min}$ in the same direction the home dummy does. If it does, the AvW resolution will turn out to be partly the literature talking itself into a smaller number using a measurement choice that did not get separately labeled. That is the next paper.

7 Limitations and future work

The paper has six limitations that a careful reader should hold in mind, and four follow-up projects that the diagnosis naturally points to.

7.1 Limitations

The empirical demonstration uses the Wei (1996) intra-national flow proxy, which is well-known to be noisy. It treats GDP minus exports as intra-national trade and is rough on services, on government

spending, and on non-traded production. The modern alternative is the Yotov-Piermartini-Monteiro-Larch (2016) “trade for production” construction from sectoral GDP. The structural critique survives under either proxy because it is about the right-hand-side distance, not the left-hand-side flow. The magnitudes of the home coefficient may shift modestly under YPMM. Running the sweep on sectoral data is a clean extension and the obvious next step.

The structural critique depends on the iceberg-cost exponent δ . I take $\delta = 1$ throughout, which is the standard gravity assumption. If you take Hummels (2007)’s freight-cost meta-evidence seriously, the freight elasticity is more like 0.2-0.4, which would put δ below 1 and the structural $\theta = \delta(1 - \sigma)$ above the pole. The pole would not bite. The paper should be read as: either the structural critique applies (under $\delta = 1$), or the literature must commit to $\delta \ll 1$ explicitly. Both are choices. The literature has been quiet about which.

The empirical sample is three years (2010, 2015, 2020). That is a triennial cross-section. A real annual panel (2000-2022, ideally with the BACI HS6 sectoral breakdown) is the next computational step. I expect the qualitative finding to survive and the standard errors to shrink.

The regularization sweep uses a uniform-disk approximation for the country geometry. Real countries are irregular, have non-uniform population density, contain internal water bodies, and so on. The country-by-country regularization curve will differ from the uniform-disk curve. The scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ is a dimension-dependent property and survives shape change. The levels will move. Doing the sweep on real country polygons via the Gonchar-Helfrich satellite-distance panel is the natural quantitative extension.

The cross-section log-OLS robustness check has small intra samples (n_{intra} between 6 and 10 across the three years). I treat this as a real limitation of that estimator on the year-by-year data, and lead with the panel PPML, which stacks the three years against 653 intra rows and absorbs the multilateral-resistance system. A Mayer-style referee will reasonably press on the cross-section’s small n , and I want the panel PPML to be the answer when they do.

Finally, the structural critique is theoretical. The empirical demonstration is one estimator family (log-OLS and PPML, two conditional means of the same kind of regression). The integral diverges as a mathematical fact, so the critique survives any estimator that uses $\log d_{ii}$ or d_{ii} on its right-hand side. Showing that the regularization sensitivity propagates into welfare counterfactuals (e.g., Caliendo-Parro 2015 exact-hat numbers for a trade-policy change) is the obvious next paper. I do not run it here.

7.2 Future work

The diagnosis points at four things to do next, and the companion satellite-distance project provides the machinery for them.

The cleanest follow-up is a regularization-aware structural estimator, the constructive version of this paper’s diagnosis. An estimator that takes $d_{\min}$ as an explicit input and returns conditional estimates would let analysts report a curve instead of a single number, with the curve’s slope itself being a quantity of interest. Paper 5b, in concept.

The next is the welfare-counterfactual extension. Caliendo and Parro’s (2015) exact-hat methodology computes the welfare effects of a trade-policy change by taking ratios of gravity-implied trade shares before and after. If the underlying d_{ii} is regularization-conditional, so are the welfare numbers. Quantifying how much depends on putting the regularization sweep through the Caliendo-Parro machinery on a real policy counterfactual (the 2021 Suez closure, Brexit, the 2018-19 US-China tariffs). The structural critique becomes a policy claim at this point.

Third is the Anderson-van Wincoop re-estimation flagged in §6. Their multilateral-resistance system is the workhorse, and it inherits the input. I conjecture the $5\times$ multiplier shrinks materially

when d_{ii} is taken seriously and may go below $1\times$ at fine spatial resolution. Verifying or falsifying the conjecture is one PPML run away.

The Gonchar-Helfrich satellite-distance panel, in flight and targeting the Journal of International Economics, provides the infrastructure for all three. The companion paper builds the bilateral effective-distance variants for 2000-2024 from WorldPop, VIIRS, and OpenStreetMap; this paper supplies the methodological language the companion panel needs to be reported honestly. The two together are the diagnosis and the empirical machine for the constructive program.

8 Conclusion

The Head-Mayer formula was always a finite object at the parameter values where Head and Mayer evaluated it. The trouble started when CEPII picked $\theta = -1$ for everyone, justified it by appeal to a reduced-form regression slope, and the literature took the resulting number as if it had been derived from the structural elasticity of substitution. It had not. The structural θ that the gravity model needs is $\delta(1 - \sigma)$, which for empirically defensible σ falls in $[-7, -3]$, entirely below the formula's pole.

CEPII's $\theta = -1$ is the harmonic-mean radial coordinate of a country, which is a perfectly defensible geometric statistic. It is not the structural CES expected pairwise distance. The two have been the same letter for twenty years, and the field has not distinguished them. The result is that every border-puzzle multiplier reported in the literature has been conditional on a measurement choice nobody discussed.

Replacing CEPII's number with a finer-resolution alternative does not fix this. It picks a different point on a curve that has no fixed point. The structurally consistent move is to acknowledge that d_{ii} does not exist as a single value at gravity-consistent σ , and to design specifications that do not require pinning it to one.

The empirical content of the diagnosis is steep. In the panel PPML the home-bias coefficient moves nearly four log units across the sweep. In the cross-section OLS it moves more than three. The 5-to-10 $\times$ border-puzzle multiplier the literature has treated as a structural fact about the world is recovered, all of it, somewhere inside the regularization window. Below the window it is a fraction of one. Above it, hundreds. The literature has been comparing apples to apples because everyone uses the same number, so the variation across studies has been small. Vary the number, even within structurally defensible bounds, and the puzzle moves with it.

Head and Mayer flagged the threads in 2014. The paper picks them up.

A Notation

A glossary of symbols used in the paper. Reduced-form θ refers to the regression coefficient of $\log X_{ij}$ on $\log d_{ij}$. Structural θ refers to the CES integration parameter $\delta(1 - \sigma)$. The two are equal only by coincidence.

B Numerical verification of Table 1

Computed in Python via direct substitution into equation (1), using `numpy.power` on `complex128` arrays for $\theta \leq -2$.

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